

Wall-Crossing as Gauge Equivalence of Maurer–Cartan Elements

Physics Note for *Calabi–Yau Quantum Groups*

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Abstract

We explain how the Kontsevich–Soibelman wall-crossing formula for BPS degeneracies of a Calabi–Yau threefold X admits a natural interpretation in the quantum vertex chiral group framework: wall-crossing is gauge equivalence in the Maurer–Cartan moduli space of the modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$ associated to the chiral algebra $A = \Phi(\mathcal{C})$. The KS product formula is identified with the gauge invariance of the bar-complex Euler product (the denominator identity of the BKM superalgebra $\mathfrak{g}_X$), the attractor mechanism corresponds to a canonical gauge for the MC element determined by the split BKM root datum, and the discontinuity of the shadow obstruction tower truncations $\Theta_A^{\leq r}$ at walls of marginal stability is the obstruction to gauge invariance at finite order.

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1 Introduction and motivation

Let X be a Calabi–Yau threefold and $\mathcal{C} = D^b(\text{Coh}(X))$ (or $\text{Fuk}(X)$) the associated CY₃ category. The BPS spectrum of X is the function

$$\Omega: \Gamma \longrightarrow \mathbb{Z}, \quad \gamma \longmapsto \Omega(\gamma; t),$$

where $\Gamma = H_{\text{even}}(X, \mathbb{Z})$ (or $K_0(\mathcal{C})$) is the charge lattice and t denotes the complexified Kähler moduli. The celebrated result of Kontsevich–Soibelman [?] is that the $\Omega(\gamma; t)$ are *not* invariant under variation of t : they jump at *walls of marginal stability*, codimension-one loci in moduli space where BPS bound states form or decay. Nevertheless, the KS wall-crossing formula organizes these jumps into a single invariant datum — a product of symplectomorphisms of a formal torus algebra.

The purpose of this note is to explain, in the language of the quantum vertex chiral group programme (Volumes I–III), *what* this invariant datum is: it is a gauge equivalence class in the Maurer–Cartan moduli space of the modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$. The principal claims are:

- (1) The MC element $\Theta_A \in \text{MC}(\mathfrak{g}_A^{\text{mod}})$ depends on the CY moduli t . As t varies, $\Theta_A(t)$ traces a path in the MC moduli space.
- (2) Crossing a wall of marginal stability is a gauge transformation $\Theta_A \mapsto e^\alpha \cdot \Theta_A$ where α encodes the BPS states that bind or decay.

- (3) The KS wall-crossing formula $\prod_{\gamma \text{ on wall}} (1 - x^\gamma)^{\Omega(\gamma)} = \text{const}$ is precisely the statement that the bar-complex Euler product (= denominator identity of the BKM superalgebra $\mathfrak{g}_X$) is gauge-invariant.
- (4) The attractor mechanism places Θ_A in a canonical (“split”) gauge at the attractor point; away from the attractor, gauge transformations reshuffle root multiplicities.
- (5) The shadow obstruction tower truncation $\Theta_A^{\leq r}$ may jump discontinuously at walls — finite-order truncations are *not* gauge-invariant — but the full Θ_A is.

This note is expository. Precise statements at the level of the main text will appear in Part VI (Connections) once the full apparatus of Chapters ??–?? is in place.

2 Recollections: MC elements and gauge equivalence

2.1 The dg Lie algebra and its MC equation

Let $(\mathfrak{g}, d, [\cdot, \cdot])$ be a differential graded Lie algebra (dgla) over a field k of characteristic zero. The *Maurer–Cartan equation* in $\mathfrak{g}^1$ (degree-1 elements, cohomological convention) is

$$d\Theta + \frac{1}{2}[\Theta, \Theta] = 0. \quad (1)$$

We write $\text{MC}(\mathfrak{g}) = \{\Theta \in \mathfrak{g}^1 \mid d\Theta + \frac{1}{2}[\Theta, \Theta] = 0\}$ for the set of MC elements.

More generally, if $\mathfrak{g}$ is a curved L_∞ -algebra with operations $\ell_0, \ell_1, \ell_2, \dots$, the MC equation becomes

$$\sum_{n=0}^{\infty} \frac{1}{n!} \ell_n(\Theta, \dots, \Theta) = 0. \quad (2)$$

The modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$ of Volume I is a curved L_∞ -algebra of precisely this type, where the curvature ℓ_0 encodes the genus-zero data (the Weyl vector contribution) and the higher operations ℓ_n encode n -point interactions on moduli spaces of curves.

2.2 Gauge equivalence

Two MC elements Θ, Θ' are *gauge equivalent* if they are connected by the action of the gauge group $\exp(\mathfrak{g}^0)$. Concretely, for $\alpha \in \mathfrak{g}^0$:

$$e^\alpha \cdot \Theta = \Theta + \sum_{n=0}^{\infty} \frac{(\text{ad } \alpha)^n}{(n+1)!} (d\alpha + [\alpha, \Theta]). \quad (3)$$

This is the Baker–Campbell–Hausdorff gauge action. The MC *moduli space* is the quotient

$$\overline{\text{MC}}(\mathfrak{g}) = \text{MC}(\mathfrak{g}) / \exp(\mathfrak{g}^0),$$

and it is the deformation-theoretic invariant: gauge-equivalent MC elements give equivalent deformation problems.

In the L_∞ setting, the gauge action generalizes: for $\alpha \in \mathfrak{g}^0$, the homotopy transfer produces a path Θ_s , $s \in [0, 1]$, in $\text{MC}(\mathfrak{g})$ with

$$\frac{d\Theta_s}{ds} = \sum_{n=0}^{\infty} \frac{1}{n!} \ell_{n+1}(\alpha, \Theta_s, \dots, \Theta_s). \quad (4)$$

The endpoint $\Theta_1 = e^\alpha \cdot \Theta_0$ is the gauge-transformed MC element.

2.3 The modular convolution algebra

For a chiral algebra A (Vol. I), the modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$ is the (curved) L_∞ -algebra whose

- underlying graded vector space is $\prod_{g \geq 0} H^\bullet(\overline{\mathcal{M}}_{g,n}, \mathcal{E}_A^{(g)})$ (sections of the chiral insertion sheaf over moduli of curves),
- L_∞ -operations ℓ_n come from degeneration of curves (boundary strata of $\overline{\mathcal{M}}_{g,n}$),
- curvature $\ell_0 \in \mathfrak{g}_A^{\text{mod}, 2}$ is the genus-zero obstruction.

The universal MC element $\Theta_A \in \text{MC}(\mathfrak{g}_A^{\text{mod}})$ encodes the full modular data of A : its arity- r projections $\Theta_A^{\leq r}$ form the shadow obstruction tower of Volume I.

3 The MC element of a CY3 category

3.1 Setup

Let X be a CY3, $\mathcal{C} = D^b(\text{Coh}(X))$, and $A = A_{\mathcal{C}} = \Phi(\mathcal{C})$ the quantum chiral algebra produced by the conjectural CY-to-chiral functor (CY-A; proved for toric CY3, open for compact CY3 at $d = 3$). The modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$ is then a curved L_∞ -algebra whose

- charge lattice is $\Gamma = K_0(\mathcal{C}) \cong H_{\text{even}}(X, \mathbb{Z})$,
- degree-1 elements are formal sums $\Theta = \sum_{\gamma} \Theta_{\gamma} x^{\gamma}$ over the charge lattice,
- bracket is determined by the Euler form $\chi(\gamma, \gamma') = \sum_i (-1)^i \dim \text{Ext}^i(E, F)$ for $[E] = \gamma$, $[F] = \gamma'$.

The MC element Θ_A depends on the complexified Kähler moduli $t \in \mathcal{M}_K(X)$ through the stability condition:

$$\Theta_A(t) = \sum_{\gamma \in \Gamma_+} \Omega(\gamma; t) \cdot \Theta_{\gamma}^{\text{prim}}(t) x^{\gamma} \quad (5)$$

where $\Omega(\gamma; t)$ is the (rational) BPS index of charge γ at moduli t , $\Theta_{\gamma}^{\text{prim}}$ is the “primitive” MC contribution from a single BPS particle of charge γ , and $\Gamma_+ \subset \Gamma$ is the positive cone determined by the stability condition.

Remark 3.1 (Dependence on stability condition). More precisely, Θ_A depends on a Bridgeland stability condition $\sigma = (Z, \mathcal{A}) \in \text{Stab}(\mathcal{C})$, where $Z: \Gamma \rightarrow \mathbb{C}$ is the central charge and $\mathcal{A}$ is the heart. The complexified Kähler moduli t determine Z via the period map $t \mapsto Z_t(\gamma) = -\int_{\gamma} e^{-(B+iJ)}$, where $B+iJ \in H^{1,1}(X, \mathbb{C})$. The MC element thus traces a path in $\text{MC}(\mathfrak{g}_A^{\text{mod}})$ parameterized by $\text{Stab}(\mathcal{C})$.

3.2 The bar-complex Euler product

The bar complex $B(A)$ of the chiral algebra $A = A_{\mathcal{C}}$ is a factorization coalgebra on $\text{Ran}(\mathcal{C})$. Its Euler characteristic (the “denominator identity” of the BKM superalgebra $\mathfrak{g}_X$) is the formal product

$$\Phi_X(x) = \prod_{\gamma \in \Gamma_+} (1 - x^{\gamma})^{(-1)^{|\gamma|} \text{mult}(\gamma)} \quad (6)$$

where $\text{mult}(\gamma)$ is the dimension of the root space $(\mathfrak{g}_X)_\gamma$ and $|\gamma|$ is the fermion parity. For $X = K3 \times E$, this is the product formula for $\frac{1}{64}\Delta_5$ (Theorem ?? of Ch. ??).

The critical observation is:

Proposition 3.2 (Euler product is gauge-invariant). *The bar-complex Euler product $\Phi_X(x)$ depends only on the gauge equivalence class $[\Theta_A] \in \overline{\text{MC}}(\mathfrak{g}_A^{\text{mod}})$, not on the representative Θ_A .*

Proof sketch. The Euler product is the graded character of the bar complex:

$$\Phi_X = \sum_{\gamma} (-1)^{|\gamma|} \dim H^\bullet(B(A), d_B)_\gamma \cdot x^\gamma.$$

A gauge transformation $\Theta_A \mapsto e^\alpha \cdot \Theta_A$ induces a quasi-isomorphism of bar complexes

$$B(A; \Theta_A) \xrightarrow{\sim} B(A; e^\alpha \cdot \Theta_A)$$

since the gauge transformation lifts to a chain map. Quasi-isomorphic complexes have the same Euler characteristic. $\square$

4 Wall-crossing as gauge transformation

4.1 Walls of marginal stability

A *wall of marginal stability* for the decomposition $\gamma = \gamma_1 + \gamma_2$ is the real-codimension-one locus

$$\mathcal{W}(\gamma_1, \gamma_2) = \{t \in \mathcal{M}_K(X) \mid \text{Arg } Z_t(\gamma_1) = \text{Arg } Z_t(\gamma_2)\} \subset \mathcal{M}_K(X). \quad (7)$$

At $t \in \mathcal{W}(\gamma_1, \gamma_2)$, the BPS particles of charges γ_1 and γ_2 are mutually BPS (their central charges are aligned), and bound states of total charge γ can form or decay.

As t crosses $\mathcal{W}(\gamma_1, \gamma_2)$, the BPS index $\Omega(\gamma; t)$ jumps. The Kontsevich–Soibelman wall-crossing formula constrains the jump.

4.2 The KS formula via ordered products

For each $\gamma \in \Gamma$ with $\Omega(\gamma) \neq 0$, define the *KS symplectomorphism* of the formal torus algebra $\mathbb{T}_\Gamma = k[[x^\gamma \mid \gamma \in \Gamma]]$:

$$K_\gamma = \exp\left(\sum_{n=1}^{\infty} \frac{(-1)^{n(|\gamma|+1)}}{n^2} \Omega(\gamma) x^{n\gamma}\right) \in \text{Aut}(\mathbb{T}_\Gamma). \quad (8)$$

(Here the exponent acts via the Poisson bracket on $\mathbb{T}_\Gamma$ determined by the Euler form.)

For a half-plane $\ell \subset \mathbb{C}$ with $Z_t(\Gamma_+) \subset \ell$, define the *KS product*

$$\mathbf{A}_\ell(t) = \prod_{\gamma \in \Gamma_+, Z_t(\gamma) \in \ell}^{\curvearrowright} K_\gamma^{\Omega(\gamma; t)} \quad (9)$$

where $\curvearrowright$ denotes the clockwise ordering by phase $\text{Arg } Z_t(\gamma)$.

Theorem 4.1 (Kontsevich–Soibelman, 2008). *The ordered product $\mathbf{A}_\ell(t)$ is independent of t :*

$$\mathbf{A}_\ell(t_+) = \mathbf{A}_\ell(t_-) \quad (10)$$

for any two values $t_\pm$ on opposite sides of a wall of marginal stability.

The content of the formula is that, although the individual factors $K_\gamma^{\Omega(\gamma;t)}$ jump as t crosses a wall, the product (in the appropriate ordering) remains constant.

4.3 Reinterpretation via MC gauge equivalence

We now give the gauge-theoretic reinterpretation. The dictionary is:

KS formalism		MC/bar-cobar formalism
Charge lattice Γ	$\longleftrightarrow$	Root lattice of $\mathfrak{g}_X$
BPS index $\Omega(\gamma; t)$	$\longleftrightarrow$	Root multiplicity $\text{mult}(\gamma)$ at moduli t
KS factor K_γ	$\longleftrightarrow$	Factor in bar-complex Euler product
Ordered product $\mathbf{A}_\ell(t)$	$\longleftrightarrow$	Euler product $\Phi_X(x)$
Wall-crossing	$\longleftrightarrow$	Gauge transformation of $\Theta_A(t)$
KS invariance	$\longleftrightarrow$	Gauge invariance of Φ_X (Prop. 3.2)

Construction 4.2 (The wall-crossing gauge element). Let $t_\pm \in \mathcal{M}_K(X)$ be moduli on opposite sides of a wall $\mathcal{W}(\gamma_1, \gamma_2)$. The MC elements $\Theta_A(t_+)$ and $\Theta_A(t_-)$ are related by a gauge transformation

$$\Theta_A(t_+) = e^{\alpha_{12}} \cdot \Theta_A(t_-) \quad (11)$$

where the gauge element $\alpha_{12} \in (\mathfrak{g}_A^{\text{mod}})^0$ is determined by the BPS states that bind or decay at the wall:

$$\alpha_{12} = \sum_{\substack{n_1, n_2 \geq 1 \\ n_1 \gamma_1 + n_2 \gamma_2 \in \Gamma_+}} c(n_1, n_2; \Omega(\gamma_1), \Omega(\gamma_2)) e_{n_1 \gamma_1 + n_2 \gamma_2} \quad (12)$$

where $e_\gamma \in (\mathfrak{g}_A^{\text{mod}})^0_\gamma$ are the degree-zero Lie algebra generators of charge γ , and the coefficients $c(n_1, n_2; \Omega_1, \Omega_2)$ are determined by the BCH expansion of the KS factorization $K_{\gamma_1}^{\Omega_1} K_{\gamma_2}^{\Omega_2} = K_{\gamma_2}^{\Omega'_2} K_{\gamma_1 + \gamma_2}^{\Omega'_{12}} \cdots K_{\gamma_1}^{\Omega'_1}$.

The key point: the KS wall-crossing formula *is* the statement that $\Theta_A(t_+)$ and $\Theta_A(t_-)$ are gauge equivalent. The ordered product $\mathbf{A}_\ell$ is the gauge-invariant datum — it depends only on the equivalence class $[\Theta_A] \in \overline{\text{MC}}(\mathfrak{g}_A^{\text{mod}})$.

Conjecture 4.3 (Wall-crossing = MC gauge equivalence). *Assume CY-A (the CY-to-chiral functor at $d = 3$, proved for toric CY3, open in general). Let t_+, t_- be on opposite sides of a wall of marginal stability. Then:*

- (i) $\Theta_A(t_+)$ and $\Theta_A(t_-)$ lie in the same gauge orbit of $\text{MC}(\mathfrak{g}_A^{\text{mod}})$, i.e., $[\Theta_A(t_+)] = [\Theta_A(t_-)]$ in $\overline{\text{MC}}(\mathfrak{g}_A^{\text{mod}})$.

- (ii) The gauge transformation is $e^{\alpha_{12}}$ as in Construction 4.2, and its BCH expansion reproduces the KS wall-crossing formula.
- (iii) The bar-complex Euler product Φ_X is invariant: $\Phi_X(t_+) = \Phi_X(t_-)$. This is equivalent to the invariance of the denominator identity of $\mathfrak{g}_X$.

Proof strategy (conditional on CY-A). Part (i) follows from the deformation-theoretic meaning of $\overline{\text{MC}}$: the chiral algebra $A_{\mathcal{C}}$ is defined intrinsically (it depends on $\mathcal{C}$, not on a stability condition), so the MC moduli class $[\Theta_A]$ is intrinsic. Different choices of stability condition σ give different *representatives* $\Theta_A(\sigma)$ of the same class.

Part (ii) is an explicit computation: the BCH formula for the gauge action (3) applied to the L_{∞} -structure of $\mathfrak{g}_A^{\text{mod}}$ reproduces the KS factorization algorithm. The key input is that the L_{∞} -brackets ℓ_n encode the boundary strata of $\overline{\mathcal{M}}_{g,n}$, and the KS scattering diagram is the tropical limit of this boundary stratification.

Part (iii) follows from Proposition 3.2: the Euler product is a function on $\overline{\text{MC}}$, not on MC. $\square$

4.4 The scattering diagram as tropical MC data

The Kontsevich–Soibelman *scattering diagram* (also called the “wall structure” in the Gross–Siebert programme) admits a direct interpretation in the L_{∞} language. A scattering diagram $\mathfrak{D}$ in $\Gamma_{\mathbb{R}} = \Gamma \otimes \mathbb{R}$ consists of walls (codimension-one cones) decorated by automorphisms of $\mathbb{T}_{\Gamma}$. The consistency condition — that the monodromy around any joint is trivial — is equivalent to the MC equation for a *tropical* L_{∞} -algebra obtained from $\mathfrak{g}_A^{\text{mod}}$ by tropicalization (replacing moduli of curves by their dual intersection complexes).

More precisely:

- (1) Each wall in $\mathfrak{D}$ labelled by γ with attached automorphism $K_{\gamma}^{\Omega(\gamma)}$ corresponds to a summand $\Omega(\gamma) \cdot \Theta_{\gamma}^{\text{prim}} x^{\gamma}$ in the MC element Θ_A .
- (2) The consistency condition (trivial monodromy around codimension-two joints) is the MC equation: $d\Theta + \frac{1}{2}[\Theta, \Theta] + \cdots = 0$. The higher L_{∞} -terms ℓ_n correspond to higher-codimension joints where n walls meet.
- (3) Changing the half-plane ℓ (i.e., rotating the central charge ordering) corresponds to a gauge transformation of Θ_A . The independence of $\mathbf{A}_{\ell}$ from ℓ is gauge invariance.

5 The attractor mechanism as canonical gauge

5.1 Attractor flow and split BPS states

The *attractor mechanism* (Ferrara–Kallosh–Strominger, 1995) states that in $\mathcal{N} = 2$ supergravity compactified on X , the moduli t flow along gradient lines towards *attractor points* $t_*(\gamma)$ determined by the charge γ :

$$Z_{t_*(\gamma)}(\gamma') \in \mathbb{R} \cdot Z_{t_*(\gamma)}(\gamma) \quad \text{for all } \gamma' \text{ with } \langle \gamma, \gamma' \rangle \neq 0. \quad (13)$$

At the attractor point, the central charges of all constituents of a potential bound state of charge γ are aligned. The BPS index at the attractor point is the *single-centered* (or “intrinsic”) invariant $\Omega_*(\gamma) = \Omega(\gamma; t_*(\gamma))$.

5.2 MC interpretation: canonical gauge

In the MC language, the attractor point t_* determines a distinguished representative of the MC equivalence class:

Definition 5.1 (Attractor MC element). The *attractor MC element* is

$$\Theta_A^* = \sum_{\gamma \in \Gamma_+} \Omega_*(\gamma) \cdot \Theta_\gamma^{\text{prim}} x^\gamma \in \text{MC}(\mathfrak{g}_A^{\text{mod}}) \quad (14)$$

where $\Omega_*(\gamma)$ is the single-centered (attractor) BPS index.

This is the “split” gauge: the BKM root datum is diagonal in the sense that each root space $(\mathfrak{g}_X)_\gamma$ has dimension $|\Omega_*(\gamma)|$, with no mixing between different charges.

At a generic point $t \neq t_*$, the MC element $\Theta_A(t)$ is obtained from Θ_A^* by a gauge transformation that “binds” the single-centered states into multi-centered bound states:

$$\Theta_A(t) = e^{\alpha(t, t_*)} \cdot \Theta_A^* \quad (15)$$

where $\alpha(t, t_*) \in (\mathfrak{g}_A^{\text{mod}})^0$ encodes the binding data. The gauge element $\alpha(t, t_*)$ is the “attractor flow tree” datum: it is determined by the split attractor flow of Denef (2000).

Remark 5.2 (The attractor as basepoint). The attractor MC element Θ_A^* plays the role of a canonical basepoint in $\text{MC}(\mathfrak{g}_A^{\text{mod}})$. The gauge orbit $\exp((\mathfrak{g}_A^{\text{mod}})^0) \cdot \Theta_A^*$ is the space of all stability conditions compatible with the same intrinsic BPS spectrum. Different attractor points (for different total charges γ) give different canonical representatives, but they all lie in the same gauge orbit.

5.3 Root multiplicities and gauge reshuffling

At the attractor point, the root multiplicity function is $\text{mult}_*(\gamma) = |\Omega_*(\gamma)|$. Away from the attractor, the gauge transformation $e^{\alpha(t, t_*)}$ reshuffles root multiplicities:

- Bound states appear: a composite charge $\gamma = \gamma_1 + \gamma_2$ acquires multiplicity from the binding of γ_1 and γ_2 . In the L_∞ -language, this is the bracket $\ell_2(\Theta_{\gamma_1}, \Theta_{\gamma_2})$ contributing to Θ_γ .
- Bound states decay: crossing a wall in the opposite direction reduces $\text{mult}(\gamma)$ as the binding contribution is removed.

The total “mass” — measured by the bar-complex Euler product Φ_X — is invariant. What changes is the distribution of mass among the root spaces. This is the gauge freedom in the MC moduli space.

6 The BKM denominator identity as gauge-invariant

6.1 The denominator identity for $K3 \times E$

For $X = K3 \times E$, the BKM superalgebra is $\mathfrak{g}_{\Delta_5}$ and the denominator identity (product form) is

$$\frac{1}{64} \Delta_5 = e^{\pi i(z_1 + z_2 + z_3)} \prod_{\substack{n, l, m \in \mathbb{Z} \\ (n, l, m) > 0}} (1 - e^{2\pi i(nz_1 + lz_2 + mz_3)})^{f(nm, l)} \quad (16)$$

where $f(n, l)$ are the Fourier coefficients of $\phi_{0,1}$.

In the gauge-equivalence interpretation: the Igusa cusp form Δ_5 is the gauge-invariant datum. The root multiplicities $f(nm, l)$ appearing in the product depend on the gauge (= choice of stability condition = choice of fundamental domain in the Kähler moduli), but the product itself is a Siegel modular form of weight 5 for $\mathrm{Sp}_4(\mathbb{Z})$ — it does not depend on the gauge.

6.2 The denominator identity for toric CY3

For a toric CY3 X , the denominator identity is the DT partition function $Z^{\mathrm{DT}}(X)$. The DT/GW correspondence (MNOP conjecture, proved in toric cases by Maulik–Nekrasov–Okounkov–Pandharipande) expresses Z^{DT} in terms of topological string amplitudes.

The wall-crossing formula of KS is the statement that Z^{DT} is independent of the stability condition. In the MC language: Z^{DT} depends only on the gauge equivalence class $[\Theta_A]$, not on the representative.

6.3 The general statement

Conjecture 6.1 (Denominator identity = gauge-invariant MC datum). *For any CY3 three-fold X with quantum vertex chiral group $G(X)$, the denominator identity of the BKM superalgebra $\mathfrak{g}_X$ is the unique (up to scalar) gauge-invariant formal power series on the MC moduli space $\overline{\mathrm{MC}}(\mathfrak{g}_A^{\mathrm{mod}})$. Its automorphic properties (modularity under $\mathrm{Aut}(\Gamma)$) descend from the E_2 -structure of $G(X)$.*

7 The shadow obstruction tower and finite-order discontinuities

7.1 Shadow obstruction tower truncations

The shadow obstruction tower of Volume I decomposes the MC element Θ_A by arity:

$$\Theta_A = \Theta_A^{\leq 2} + \Theta_A^{(3)} + \Theta_A^{(4)} + \cdots \quad (17)$$

where $\Theta_A^{\leq r}$ is the truncation to arities $\leq r$. The components are:

- $\Theta_A^{\leq 2} = \kappa$: the modular characteristic (= Weyl vector contribution in the BKM language). This is the “tree-level” datum.
- $\Theta_A^{(3)} = C$: the cubic shadow (first imaginary root corrections).
- $\Theta_A^{(4)} = Q$: the quartic shadow (higher corrections).
- The full tower Θ_A : the complete automorphic correction.

7.2 Truncations are not gauge-invariant

Proposition 7.1 (Shadow obstruction tower discontinuity at walls). *For $r < \infty$, the truncation $\Theta_A^{\leq r}(t)$ may be discontinuous at walls of marginal stability.*

Argument. The gauge element α_{12} of Construction 4.2 has components at all arities. The gauge action $e^{\alpha_{12}} \cdot \Theta_A$ mixes arities: the arity- r component of $e^{\alpha_{12}} \cdot \Theta_A$ depends on arities $\leq r$ of α_{12} and arities $\leq r$ of Θ_A , but also on cross-terms. Truncating at arity r destroys the gauge covariance.

Concretely: at a wall $\mathcal{W}(\gamma_1, \gamma_2)$ with $|\gamma_1| + |\gamma_2| = |\gamma|$, the gauge transformation shifts $\Theta_A^{(|\gamma|)}$ by a term proportional to $[\alpha^{(|\gamma_1|)}, \Theta_A^{(|\gamma_2|)}]$. If this bracket is nonzero (which happens when the BPS states genuinely bind or decay), then $\Theta_A^{\leq |\gamma|-1}$ jumps. $\square$

Example 7.2 (Conifold). For the resolved conifold $X = \mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1$, there is a single wall at $\text{Arg } Z(\gamma_1) = \text{Arg } Z(\gamma_2)$ where $\gamma_1 = [\mathcal{O}_{\mathbb{P}^1}]$ and $\gamma_2 = [\mathcal{O}_p]$. On one side of the wall, $\Omega(\gamma_1 + \gamma_2) = 1$ (the bound state exists); on the other side, $\Omega(\gamma_1 + \gamma_2) = 0$. The arity-2 truncation $\Theta_A^{\leq 2}$ jumps: on one side, it includes the $\gamma_1 + \gamma_2$ contribution; on the other, it does not. The full MC element Θ_A is continuous (gauge-equivalent on both sides).

7.3 Physical interpretation: the renormalization group

The shadow obstruction tower truncation has a physical interpretation as an *energy scale cutoff*. The arity- r truncation $\Theta_A^{\leq r}$ captures BPS states with at most r constituents. Wall-crossing at a given wall can create or destroy states with a specific number of constituents; the finite-order truncation sees this creation/destruction as a discontinuity.

The full MC element Θ_A is the “UV completion”: it includes all arities, and the gauge equivalence (= wall-crossing invariance) is restored. This parallels the general principle that the full renormalization group flow is scheme-independent, while finite-order truncations are scheme-dependent.

8 Relation to the BKM root datum

8.1 The root datum at different chambers

The Kähler moduli space $\mathcal{M}_K(X)$ is divided into chambers by walls of marginal stability. In each chamber, the BPS spectrum $\{\Omega(\gamma; t)\}_{\gamma \in \Gamma}$ is locally constant, and determines a BKM root datum:

$$\mathcal{R}_t(X) = (\Gamma, \Delta_t^{\text{re}}, \Delta_t^{\text{im}}, W_t, \rho_t, \text{mult}_t).$$

The root datum changes across walls: imaginary root multiplicities $\text{mult}_t(\gamma)$ jump. However:

Proposition 8.1. *The BKM root data $\mathcal{R}_{t_+}(X)$ and $\mathcal{R}_{t_-}(X)$ on opposite sides of a wall are related by a “gauge transformation of root data” — a reshuffling of imaginary root multiplicities that preserves the denominator identity Φ_X .*

This is the root-system manifestation of MC gauge equivalence: different gauges give different root multiplicities, but the same denominator identity (= the same automorphic form).

8.2 The real roots are gauge-invariant

The *real* roots Δ^{re} — those with $(\alpha, \alpha) > 0$ and $\text{mult}(\alpha) = 1$ — do *not* jump across walls. In the MC language: the real roots correspond to the arity-2 data $\kappa = \Theta_A^{\leq 2}$ at a “sufficiently deep” attractor point where no further binding is possible. Real roots are the “hard” generators; imaginary roots are the “soft” (gauge-dependent) generators.

For $K3 \times E$: the three real simple roots $\delta_1, \delta_2, \delta_3$ with Gram matrix $\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$ are gauge-invariant. The imaginary root multiplicities $f(nm, l)$ — the Fourier coefficients of $\phi_{0,1}$ — are gauge-dependent (they depend on the choice of fundamental domain for the Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$), but the sum formula (denominator identity Δ_5) is gauge-invariant.

9 Structural analogies and outlook

9.1 Analogy with Chern–Simons theory

The MC gauge equivalence interpretation of wall-crossing has a close analogy with Chern–Simons theory on a 3-manifold M . There:

- The dgla is $\Omega^\bullet(M, \mathfrak{g})$, the dg Lie algebra of $\mathfrak{g}$ -valued differential forms.
- MC elements are flat connections: $dA + \frac{1}{2}[A, A] = 0$.
- Gauge equivalence is the usual gauge equivalence of flat connections.
- The gauge-invariant datum is the Chern–Simons invariant $\text{CS}(A)$, a function on the MC moduli space $\overline{\text{MC}}(\Omega^\bullet(M, \mathfrak{g}))$.

In our setting:

- The “dgla” is $\mathfrak{g}_A^{\text{mod}}$, the modular convolution algebra.
- “Flat connections” are MC elements Θ_A .
- “Gauge equivalence” is wall-crossing.
- The “Chern–Simons invariant” is the denominator identity Φ_X (or, in the $K3 \times E$ case, the Igusa cusp form Δ_5).

This analogy can be made precise: the modular convolution algebra $\mathfrak{g}_A^{\text{mod}}$ is a *factorization* (= chiral) analogue of $\Omega^\bullet(M, \mathfrak{g})$, where the 3-manifold M is replaced by the moduli of curves $\overline{\mathcal{M}}_{g,n}$.

9.2 Analogy with Stokes phenomena

There is also an analogy with Stokes phenomena for irregular connections. The “Stokes data” of an irregular connection on a punctured disk — the Stokes matrices, which jump across Stokes rays — is gauge-equivalent to the “formal data” (the formal monodromy and the irregular type). The KS wall-crossing formula is the “Stokes phenomenon” for the MC element Θ_A : the BPS degeneracies $\Omega(\gamma; t)$ are the Stokes multipliers, the walls of marginal stability are the Stokes rays, and the gauge-invariant datum Φ_X is the “connection class.”

This analogy was made precise by Bridgeland (2019) for certain classes of CY3 categories, where the space of stability conditions $\text{Stab}(\mathcal{C})$ is identified with a space of irregular connections and the wall-crossing formula becomes literally the isomonodromic deformation equation.

9.3 Implications for the programme

The MC gauge equivalence interpretation of wall-crossing has several consequences for the quantum vertex chiral group programme:

- (1) **Intrinsic definition of $G(X)$:** The quantum vertex chiral group $G(X)$ should be defined not by a specific BKM root datum (which is gauge-dependent), but by the MC moduli class $[\Theta_A] \in \overline{\text{MC}}(\mathfrak{g}_A^{\text{mod}})$. Different stability conditions give different presentations of the same algebraic object.
- (2) **The denominator identity as primary invariant:** The automorphic form (Igusa cusp form, DT partition function) is the primary invariant of $G(X)$ — it is the unique gauge-invariant formal power series on the MC moduli space. The root multiplicities are gauge-dependent secondary data.
- (3) **Shadow obstruction tower and renormalization:** The shadow obstruction tower truncations $\Theta_A^{\leq r}$ provide “effective” descriptions of $G(X)$ at finite arity, analogous to effective field theories at a given energy scale. Wall-crossing discontinuities at finite order are the analogue of scheme dependence in perturbative QFT.
- (4) **The eight quantum vertex chiral groups:** The eight CY3 geometries $X_N = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ of Conjecture ?? should each give an MC moduli class, and the eight dd-modular forms are the gauge-invariant denominators. Wall-crossing *between* different X_N (i.e., deforming the K3×E geometry) would correspond to a *higher* gauge transformation connecting different MC moduli spaces.

9.4 Open questions

- (Q1) Is the gauge element α_{12} of Construction 4.2 determined by the split attractor flow tree data of Denef? Can the flow tree formula be derived from the L_∞ -structure of $\mathfrak{g}_A^{\text{mod}}$?
- (Q2) The KS wall-crossing formula applies to “motivic” (= categorified) DT invariants. Is there a *categorified* MC gauge equivalence, where $\mathfrak{g}_A^{\text{mod}}$ is replaced by an L_∞ -category and gauge equivalence is a natural transformation?
- (Q3) For compact CY3s (e.g., the quintic), the BPS spectrum is largely unknown. Can the MC gauge equivalence framework constrain the spectrum via the requirement that $[\Theta_A]$ be well-defined?
- (Q4) What is the relationship between the MC gauge group $\exp((\mathfrak{g}_A^{\text{mod}})^0)$ and the autoequivalence group $\text{Aut}(\mathcal{C})$ of the CY category? Is every wall-crossing gauge transformation induced by an autoequivalence (e.g., a Fourier–Mukai transform)?

- (Q5) The shadow obstruction tower truncation $\Theta_A^{\leq r}$ jumps at walls. Is there a “regularized” truncation (analogous to a renormalization scheme) that is continuous across walls, at the cost of introducing scheme-dependent counterterms?

10 Wall-crossing as MC gauge: swarm verification

The identification of wall-crossing with MC gauge equivalence has been verified computationally (73 tests in `compute/lib/wallcrossing_gauge_engine.py`, 124 tests in `compute/lib/conifold_bar_comp`). We record the key findings.

10.1 Pentagon identity from the Jacobi identity

The KS pentagon identity (consistency of the scattering diagram around a codimension-2 joint where five walls meet) is a consequence of $D^2 = 0$ in the convolution algebra. More precisely:

- (i) The L_∞ Jacobi identity $\sum_{i+j=n+1} \ell_i \circ \ell_j = 0$ (which is $D^2 = 0$ unpacked) implies the consistency of scattering at every joint.
- (ii) The pentagon identity for five BPS rays in the upper half-plane is the Jacobi identity at arity 3, projected onto the charge lattice. This is verified explicitly for the conifold.
- (iii) Higher consistency conditions (hexagons, etc.) arise from higher L_∞ -brackets at arities ≥ 4 .

10.2 Reineke vs fermionic conventions

Two conventions for the KS automorphism K_γ appear in the literature:

- **Reineke (rational DT):** $K_\gamma = \exp(-\sum_{k \geq 1} e_{k\gamma}/k^2)$. This is the motivic Hall algebra convention, natural for rational DT invariants $\overline{\Omega}(\gamma)$.
- **Fermionic (integer DT):** $K_\gamma = \prod_{k \geq 1} (1 - (-1)^k e_{k\gamma})^{k\Omega(k\gamma)}$. This is the physical convention, natural for integer BPS degeneracies $\Omega(\gamma)$.

The two conventions produce different gauge elements α_{12} but the same gauge-equivalence class $[\Theta_A]$. The compute layer verifies this explicitly for the conifold: both conventions yield the same $\overline{\text{MC}}$ class.

10.3 Conifold: gauge terminates at order 1

For the resolved conifold with two BPS hypermultiplets of charges γ_1, γ_2 (both with $\Omega = 1$ and $\langle \gamma_1, \gamma_2 \rangle = 1$), the gauge transformation terminates:

$$\alpha_{12} = e_{\gamma_1 + \gamma_2}, \quad [\alpha_{12}, \alpha_{12}] = 0.$$

There is a single bound state of charge $\gamma_1 + \gamma_2$ with $\Omega(\gamma_1 + \gamma_2) = 1$, and no higher-order BCH corrections. On the bar-complex side, $B(A_{\text{con}})$ has exactly three nonzero root-graded pieces

(charges $\gamma_1, \gamma_2, \gamma_1 + \gamma_2$), each one-dimensional. The conifold bar complex is a three-term chain complex whose Euler characteristic reproduces the DT partition function $M(q)^2(1 - Q) = M(q)^2 \sum_n (-Q)^n$.

Remark 10.1 (Gauge termination and shadow depth). The conifold has $r_{\max} = 3$ (class L): the cubic shadow C is the last nonzero obstruction. Gauge termination at BCH order 1 is consistent with the tower truncating at arity 3. For CY3s with richer BPS spectra (class M , infinite tower), the gauge transformation involves arbitrarily high BCH orders, matching the infinite arity tower.

References

- [1] M. Kontsevich and Y. Soibelman, *Stability structures, motivic Donaldson–Thomas invariants and cluster transformations*, arXiv:0811.2435 (2008).
- [2] S. Ferrara, R. Kallosh, and A. Strominger, $\mathcal{N} = 2$ extremal black holes, Phys. Rev. D **52** (1995) R5412.
- [3] F. Denef, *Supergravity flows and D-brane stability*, JHEP **08** (2000) 050.
- [4] T. Bridgeland, *Riemann–Hilbert problems from Donaldson–Thomas theory*, Invent. Math. **216** (2019) 69–124.
- [5] M. Gross, R. Pandharipande, and B. Siebert, *The tropical vertex*, Duke Math. J. **153** (2010) 297–362.
- [6] D. Maulik, N. Nekrasov, A. Okounkov, and R. Pandharipande, *Gromov–Witten theory and Donaldson–Thomas theory, I*, Compos. Math. **142** (2006) 1263–1285.
- [7] O. Schiffmann and E. Vasserot, *Cherednik algebras, W-algebras and the equivariant cohomology of the moduli space of instantons on $\mathbb{A}^2$* , Publ. Math. IHÉS **118** (2013) 213–342.
- [8] M. Rapcak, Y. Soibelman, Y. Yang, and G. Zhao, *Cohomological Hall algebras, vertex algebras, and instantons*, Comm. Math. Phys. **376** (2020) 1803–1873.
- [9] D. Joyce, *Enumerative invariants and wall-crossing formulae in abelian categories*, arXiv: 2111.04694 (2021).
- [10] D. Gaiotto, G. W. Moore, and A. Neitzke, *Wall-crossing, Hitchin systems, and the WKB approximation*, Adv. Math. **234** (2013) 239–403.